RES POLYTECHNIC, BANGALORE-91

CIE Practice Test IV

Program	Computer Science and Engineering		Semester- III
Course Name	Data Structure with Python		Test IV
Course Code	25CS33I	Duration- 180 min	Marks-50
Name of the Course Coordinator: Vinay Kumar			
Note: Answer any one full question from each section. Each full question carries equal marks. Write the program, execute it and show the output to the examiner.			

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	**a)** Write a Python program to implement a Stack using a class (LIFO principle) with the following operations: **(12 marks)** 1. push(item), pop() and peek() with a message when the stack is empty. 2. is_empty() and display(). Push the elements 10, 20, 30, 40, perform two pop operations and display the remaining stack. **b)** Write a menu-driven Python program using the Stack class to show a real-life example of a stack (a pile of plates in a canteen): **(13 marks)** 1. Add a plate (push), take a plate (pop) and see the top plate (peek). 2. Display all the plates and display a message when there are no plates.	L3	1, 3	25
	OR			
2	**a)** Write a Python program to implement a Queue using a class (FIFO principle) with the following operations: **(12 marks)** 1. enqueue(item) and dequeue() with a message when the queue is empty. 2. is_empty() and display(). Enqueue 10, 20, 30, 40, perform two dequeue operations and display the remaining queue. **b)** Write a menu-driven Python program using the Queue class to show a real-life example of a queue (customers at a ticket counter): **(13 marks)** 1. Add a customer (enqueue) and serve a customer (dequeue). 2. Display the waiting customers and display a message when no customer is waiting.	L3	1, 3	25

Q. No	Questions	CL	CO	Marks
	SECTION II			
3	**a)** Write a Python program to implement a Singly Linked List with a Node class and the following operations: **(12 marks)** 1. Insertion at the beginning, at the end and in the middle. 2. Deletion from the beginning, from the end and from the middle. 3. Traversal and searching for a value. **b)** Write a Python program to implement a Doubly Linked List (with previous and next pointers) and the following operations: **(13 marks)** 1. Insertion at the beginning, at the end and in the middle. 2. Deletion from the beginning, from the end and from the middle. 3. Traversal in the forward direction and in the backward direction.	L3	1, 3, 4, 5	25
	OR			
4	**a)** Write a Python program to create a Binary Tree using a Node class by inserting the values 50, 30, 70, 20, 40, 60, 80 and: **(12 marks)** 1. Search for a value entered by the user and display whether it is found. 2. Delete a leaf node entered by the user. 3. Display the In-order, Pre-order and Post-order traversals using recursion. **b)** For the tree created in (a), write a Python program to: **(13 marks)** 1. Display the In-order, Pre-order and Post-order traversals using an iterative method (use a stack). 2. Compare the iterative output with the recursive output.	L3	1, 3, 4, 5	25

Note for the Course Coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.

Sign of the Course Coordinator Signature of the HOD Signature of the IQAC Chairman